

NAME: _____

Teacher: _____

Multiple Choice Booklet Number: _____



SHENTON
COLLEGE

A1/A2 CHEMISTRY - TEST 1 (2017)

TIME: 50 Minutes

Properties and Structure of Atoms and Bonding

PUT A CROSS (X) THROUGH THE CORRECT ANSWER.

1.	a	b	c	d
2.	a	b	c	d
3.	a	b	c	d
4.	a	b	c	d
5.	a	b	c	d
6.	a	b	c	d
7.	a	b	c	d
8.	a	b	c	d

9.	a	b	c	d
10.	a	b	c	d
11.	a	b	c	d
12.	a	b	c	d
13.	a	b	c	d
14.	a	b	c	d
15.	a	b	c	d

PART A /15	
PART B /35	
TOTAL /50	
PERCENTAGE	

PART B: WRITTEN SECTION

1. Calculate the number of protons, neutrons and electrons in each of the following; the number represents the mass number of this particle.

Atom/ion	Number of protons	Number of electrons	Number of neutrons
Sodium-23 ion			
Arsenic-74 atom			
Oxide-18 ion			
Neon-20 atom			

(4 marks)

2. Classify the following substances as elements, compounds or mixtures. If it is a mixture state whether it is homogeneous or heterogeneous.

SUBSTANCE	ELEMENT, COMPOUND, MIXTURE	HOMOGENEOUS / HETEROGENEOUS
Solid copper oxide		
A cordial solution		
Chocolate chip biscuit		
Nitrogen gas		

(3 marks)

3. Use the following information to answer the questions below about atoms of elements A, B, C, D and E. **(5 marks)**

ELEMENT	ATOMIC NUMBER	MASS NUMBER	ELECTRONIC STRUCTURE
A	3	7	2,1
B	17	36	2,8,7
C	10	21	2,8
D	12	24	2,8,2
E	19	39	2,8,8,1

- Which element has 20 neutrons in each atom? _____ (1)
- Which two elements are in the same Group of the Periodic Table?
_____ & _____ (1)
- Which two elements **form ions** with the same electronic structure as argon?
_____ & _____ (1)
- The above mass number of Element D is not rounded off and the relative atomic mass is listed as 24.3 on the Periodic Table. Why is the relative atomic mass different to the mass number in this question? (2)

4. Calculate the relative atomic mass of a sample of lithium, given that a mass spectrometer shows that it consists of 7.40% of ${}^6\text{Li}$ and 92.6% of ${}^7\text{Li}$. Show all your working. **(3 marks)**

5. Draw electron dot diagrams of the following substances.

(8 marks)

a. CO_2

b. Na_3P

c. NF_3

d) Na_2CO_3

6. Suggest a property that could be used to distinguish between each of the following different types of solid. In each case, give a detailed explanation for your choice, linking the property to the structure of the materials. **(6 marks)**

a. An ionic solid and a metal.

Property

Reason

- b. A covalent network solid and a molecular solid.

Property

Reason

(3)

7. Give two reasons why atomic absorption spectroscopy (AAS) **might** be used instead of a flame test.

I) _____

II) _____

(2 marks)

8. (4 marks)

- a. Outline the steps you would take to obtain each **solid** separately from a mixture of sand and copper sulfate. Include what will happen in each step. (3)

- b. Which physical properties (at least 2) are being used to separate the mixture?

(1)

Written Section Total

/35